Lab 4: Bivariate Relationships and Basic Statistical Tests

For this session, we will use data from the NSECE 2019 Workforce Questionnaire to examine the relationship between two variables and perform basic statistical tests to determine whether the observed differences or associations are statistically significant.

1. Set up your working environment

Load the following packages: tidyverse, ggplot2.

2. Prepare the data

Subset the NSECE data to keep the following variables:

- WF9_WORK_FT (staff full-time or part-time status)
- WF9_CHAR_GENDER (staff gender)
- WF9_WORK_MONTHS (number of months worked in the past 12 months)
- WF9_WORK_RESPECT (agreement with: "my co-workers and I are treated with respect")
- WF9_WORK_WAGE (hourly wage)

Rename these variables to more readable names. Perform any data cleaning you find necessary. For each variable, indicate its type (nominal, ordinal, interval/ratio; discrete or continuous).

3. Bivariate relationships

- Examine the relationship between **employment status** (full-time vs. part-time) and **gender** among the child care workforce employed in center-based programs across the United States.
- Examine the relationship between **hourly wage** and **gender** among the child care workforce employed in center-based programs across the United States.
- Examine the relationship between **hourly wage** and the **number of months worked during the past 12 months**. In addition to visualizing the relationship, calculate and interpret the correlation coefficient between these two variables.

4. Statistically significant differences and associations

For each of the relationships examined above, conduct the appropriate statistical test to determine whether the observed differences or associations are statistically significant. Interpret the results and explain your conclusions.